

Recap

Monitoring refers to the collection of measurements

Methodological approach consisting of interdependent steps

Solid domain knowledge & multidisciplinary competences

Active monitoring: real technological infrastructure & controlled artificial load/traffic

ping is a very simple traffic generator based on ICMP protocol used to measure, test and debug network configurations

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18

ping limitations

There are several *limitations* affecting the accuracy of the measurements because the network is seen as “black-box”

- Impossible to understand why the target is unreachable (e.g., network “broken”, target down, intermediate routers discarding packets)
- Impossible to detect network congestions (i.e., whether and where)

To ensure measurement quality, it is important to repeat the tests multiple times especially in the case of shared media

These tests should not affect the normal operations of the networks, e.g., avoid working hours, or using automated scripts because of the traffic being generated (e.g., flooding)

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Active monitoring: **traceroute**

traceroute is a simple active monitoring tool aimed at discovering the network topology between a vantage point and a target

The tool leverages the Time To Live (TTL) header field of IP packets

The traffic generated from the vantage point consists of (multiple) *independent* sequences of *independent* packets (by default three) towards the target (implementations based on ICMP, TCP or UDP protocols)

Each sequence is characterized by a different value of TTL (i.e., 1, 2, ..., N)

A TTL equal to zero elicits an ICMP `TIME_EXCEEDED` response from the routers along the path to the target back to the vantage point

The target generates an ICMP *“port unreachable”* response

Target and intermediate routers are passive entities

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20

Active monitoring: **traceroute** scenario

Why:

- Network troubleshooting: discover network topology, loops, misconfigurations, malfunctioning routers

Which:

- Server B

What:

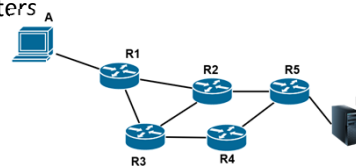
- Path between vantage point and target
- Instantaneous RTTs between vantage point and all intermediate routers
- Instantaneous RTTs between vantage point and target

Where:

- Computer A

How:

- **traceroute**

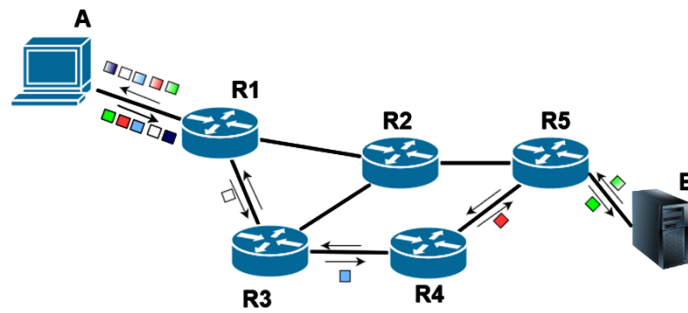


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Implementation



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traceroute limitations

There are several *limitations* affecting the accuracy of the measurements

- Asymmetric paths (e.g., network congestions)
- Unstable paths (e.g., temporary failures)
- Blocked paths (e.g., disconnected networks, routers discarding packets)

Packets are independent entities that might follow different paths

Network providers might hide their internal structure, thus leading to “hidden hops”

It is important to *repeat* the tests multiple times and discard inaccurate measurements to avoid wrong conclusions

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Stress test tools

Stress test tools are load/traffic generators aimed at measuring the (maximum) performance of services and infrastructure under a given load

These tools can be used for malicious purposes (e.g., stressers/booters used by DDoS-for-hire services)

Speed testing tools are a specific type of stress test tools

Speed testing tools aim at measuring the network speed by generating *artificial* traffic towards a server chosen as a target

Typical “black-box” approach

The traffic consists of downloads, uploads, pings and ...

In general, measurements being collected refer to downstream and upstream throughput, latency, jitter, at a given instant of time

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24

Speed testing tool limitations

Speed testing tools are affected by several limitations:

- Measurements are influenced by many factors, e.g., testing infrastructure, user hardware/software, “distance” between vantage point and target, network conditions
- Artificial traffic being generated is not always representative of Internet experience of the users

To improve the accuracy of the measurements, it is necessary to repeat the tests multiple times at different times of the day and in different days of the week

It is also important to vary the target

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25

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26

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27